Effective-sample-size calibration of a modality test for selecting the number of clusters in spatially autocorrelated data

F. Parra [co-authors]

Draft — June 5, 2026

Abstract

Selecting the number of clusters K is a basic step in unsupervised classification. The standard internal criteria — the gap statistic, the average silhouette, the Calinski–Harabasz and Davies–Bouldin indices — assume independent observations and are therefore miscalibrated on spatial data, where neighbouring locations are autocorrelated and the effective sample size is far below the nominal one, and where a smooth large-scale trend mimics discrete structure. We show that the consequence is systematic over-detection: on autocorrelated fields without discrete classes these criteria report spurious clusters in every one of 75 controlled realisations, and on real imagery they fragment single land-cover regions. We recast cluster-number selection as a test for *multimodality in excess of spatial autocorrelation*. The proposed method, ESS-Dip, evaluates the Hartigan dip statistic along the locally optimal split direction and calibrates it against the effective sample size derived from a within-class autocorrelation range, which a local declustering estimator separates from the between-class scale; a low-order trend is removed adaptively. A recursive partitioning returns K, and the test carries an asymptotic false-split-control property. On a factorial simulation with known ground truth, ESS-Dip is the only criterion among thirteen to combine perfect specificity with competitive exact-K recovery and attains the best balanced score; on the Indian Pines, Salinas and Sentinel-2 scenes it identifies single land-cover regions as homogeneous where the standard criteria do not. The method is conservative by design — a guard against the over-segmentation the standard criteria exhibit rather than an exhaustive enumerator of weakly separated or strongly imbalanced classes, a trade-off we characterise. Code and data are released.

Keywords: number of clusters · cluster validity · spatial autocorrelation · effective sample size · declustering · unsupervised classification · remote sensing

1 Introduction

Unsupervised classification partitions a data set into groups without labelled examples, and in doing so it must answer a prior question: how many groups are there? The choice of the number of clusters K governs every downstream interpretation, and in the analysis of remotely sensed imagery — where K is the number of land-cover classes an analyst will map — it is made routinely and consequentially. A mature toolkit exists to make it: the gap statistic (Tibshirani et al., 2001), the average silhouette (Rousseeuw, 1987), the Calinski–Harabasz (Caliński and Harabasz, 1974) and Davies–Bouldin (Davies and Bouldin, 1979) indices each score a partition by its internal geometry and select the K that optimises the score.

These criteria share a hidden assumption: that the observations are independent. Their reference levels — the uniform null of the gap statistic, the implicit baseline of a silhouette — describe what the geometry would look like were the samples exchangeable draws. Spatial data are not exchangeable. Neighbouring pixels are autocorrelated (Tobler, 1970), so the effective number of independent observations is far below the nominal pixel count (Cressie, 1993), and a smooth illumination or topographic gradient spreads the feature cloud along a manifold that an independence null cannot distinguish from genuine groups. The consequence, which we quantify in Sections 4–5, is systematic *over-detection*: on synthetic fields containing no discrete classes the gap statistic reports on average between seven and eight, and on real imagery the silhouette, Calinski–Harabasz and Davies–Bouldin indices fragment a single, homogeneous crop field into several sub-classes in every instance we examined. A criterion that cannot recognise the absence of clusters is a precarious foundation for deciding their number.

Correcting this is not a matter of swapping in a better null. Because the within-cluster dispersion that these criteria minimise is a second-order quantity, a reference matched to the data's second-order spatial structure — its variogram — reproduces the same dispersion and explains the clusters away. Distinguishing real classes from autocorrelation therefore requires information beyond second order, namely the *multimodality* of the feature distribution; and distinguishing real classes from a smooth trend requires separating a continuous drift from discrete structure, a separation the geostatistical decomposition into trend and residual makes natural. The idea of calibrating a clustering criterion against an autocorrelation-aware null is itself not new — Hennig and Lin (2015) do exactly this for discrete presence–absence data — but it has not been carried to the continuous-field setting of gridded imagery, nor has the confounding of trend with class been addressed in selecting K.

This paper supplies both. We recast cluster-number selection as a test for *multimodality in excess of spatial autocorrelation* and make it valid on autocorrelated fields by calibrating the test at the effective sample size. The construction rests on three geostatistical devices: the effective sample size itself (Cressie, 1993; Dutilleul, 1993), which sets the calibration; a *local declustering* estimate of the within-class autocorrelation range (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998), which prevents the between-class mosaic from corrupting that calibration; and an adaptive trend–residual decomposition, which removes a smooth drift only when it is statistically resolvable. A recursive partitioning driven by the calibrated dip statistic (Hartigan and Hartigan, 1985) returns K. Our contributions are:

- We document, on controlled ground truth, that the standard number-of-clusters criteria over-detect under spatial autocorrelation without exception, failing to recognise class-free scenes in every one of 75 realisations (Section 4).
- We propose *ESS-Dip*, an effective-sample-size-calibrated modality test for K, combining a within-class range from local declustering with adaptive trend removal (Section 3), and state an asymptotic false-split-control property (Property 1) that the method confirms empirically.
- We evaluate thirteen criteria on a factorial simulation with known K, including the calibrated-validity-index method of Hennig and Lin (2015) that ESS-Dip generalises; ESS-Dip is the only method to combine perfect specificity with competitive recovery, attaining the best balanced score, and an ablation attributes the gain to the local range and the adaptive detrending (Section 4).
- We confirm the behaviour on three real scenes spanning hyperspectral and multispectral sensing — Indian Pines, Salinas, and a Sentinel-2 scene labelled by ESA WorldCover — where ESS-Dip identifies single land-cover regions as homogeneous where the standard criteria do not (Section 5).

The remainder of the paper is organised as follows. Section 2 reviews number-of-clusters criteria, the geostatistical notions of effective sample size and declustering, and modality-based clustering, and positions the contribution. Section 3 develops the method and its calibration. Sections 4 and 5 report the simulation study and the application to land-cover imagery. Section 6 discusses the precision–recall trade-off and limitations, and Section 7 concludes. Code and data are released to reproduce every result.

2 Background and related work

2.1 Selecting the number of clusters

Choosing the number of clusters K is a classical problem in unsupervised analysis, and a large family of *internal* criteria address it by scoring a partition against the geometry of the data alone. The gap statistic (Tibshirani et al., 2001) compares the within-cluster dispersion to its expectation under a reference distribution with no clustering, typically uniform over the data's bounding box; the average silhouette (Rousseeuw, 1987), the Calinski–Harabasz ratio (Caliński and Harabasz, 1974) and the Davies–Bouldin index (Davies and Bouldin, 1979) balance within- against between-cluster scatter. Common to all is the assumption that the observations are exchangeable: the reference distribution, or the index's implicit baseline, treats the samples as independent draws. When this holds the criteria are well calibrated; when it fails they are not. Spatial data are the archetypal failure, because nearby locations are correlated — "everything is related to everything else, but near things are more related than distant things" (Tobler, 1970).

2.2 Accounting for spatial dependence

That an independence reference over-detects structure in dependent data is known across several fields. In spatial epidemiology, Goovaerts and Jacquez (2004) show that a complete-spatial-randomness null over-identifies disease clusters and propose, in its place, a geostatistical *spatial neutral model* — realisations reproducing the histogram and variogram of the data by sequential Gaussian simulation — against which far fewer locations remain significant. In neuroimaging, Eklund et al. (2016) trace inflated false-positive rates in cluster-extent inference to a mis-specified parametric model of the spatial autocorrelation. The common remedy is a null that *preserves* the second-order spatial structure while removing the effect being tested, realised either by geostatistical simulation (Goovaerts and Jacquez, 2004) or by spatially constrained randomisation such as Moran spectral randomization (Wagner and Dray, 2015).

The same logic has been brought to bear on K itself. Hennig and Lin (2015) calibrate a validity index (the gap, silhouette or prediction strength) against its distribution under a problem-specific null that encodes non-clustering structure, including spatial autocorrelation, and use the calibrated index both to test for clustering and to estimate K; on a biogeographic example a plain Gaussian null favours eight clusters while the autocorrelation-aware null reduces this to two and renders the apparent clustering non-significant. Their framework is the conceptual basis of the present work. It is, however, formulated for discrete presence–absence data with an algorithmic disjunction-parameter null, and addresses neither the *continuous-field* setting of gridded imagery nor the confounding of a smooth large-scale *trend* with discrete classes. The present contribution supplies exactly these: a Gaussian-random-field null on the lattice, a within-class range obtained by local declustering, an adaptive trend–residual separation, and an effective-sample-size calibration of a modality statistic.

The quantitative device that makes this possible is the *effective sample size*: under spatial autocorrelation the variance of a sample statistic behaves as if computed from $n_{\text{eff}} < n$ independent observations, a reduction governed by the integral of the autocorrelation (Cressie, 1993). Dutilleul (1993) operationalised this idea to correct the degrees of freedom of a correlation test between two spatial processes; we use it to recalibrate, rather than to correct after the fact, the reference distribution of a clustering criterion. The autocorrelation scale it requires is the variogram range (Matheron, 1963; Cressie, 1993), and the safeguard against letting large homogeneous regions dominate its estimate is *declustering*, a standard geostatistical correction for spatially clustered support (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998).

2.3 Modality as a clustering signal

Because k-means dispersion is itself a second-order quantity, a null matched to the data's second-order structure can explain away the very clusters one seeks; a criterion that survives such a null must use information beyond second order. Multimodality is the natural choice. The dip test of Hartigan and Hartigan (1985) measures the departure of a univariate sample from unimodality, and dip-means (Kalogeratos and Likas, 2012) turns it into a divisive procedure that splits a cluster only when a projection of it is significantly multimodal, thereby estimating K. These methods assume independent observations; our contribution is to make the dip test valid under spatial autocorrelation by calibrating it at the effective sample size, and to pair it with the trend removal that prevents a smooth gradient from being read as two modes.

2.4 Distinction from spatially constrained clustering

A separate line of work *imposes* spatial contiguity on the partition: regionalization methods such as SKATER (Assunção et al., 2006), ClustGeo (Chavent et al., 2018) and the max-p-regions problem (Duque et al., 2012) constrain clusters to be spatially connected. These address a different question — how to partition space into contiguous regions — and still require the number of regions as an input or a tuning parameter. We do not constrain the partition; we correct the criterion that selects *how many* groups are present, and the two are complementary: the effective-sample-size calibration developed here could in principle inform the stopping rule of a regionalization just as it informs the recursion of Section 3.

3 Mathematical formulation

We formulate the selection of the number of clusters as a sequence of hypothesis tests for *multimodality in excess of spatial autocorrelation.* Section 3.1 fixes notation; Section 3.2 states the null against which a candidate split is judged; Sections 3.3–3.6 develop the test statistic, its effective-sample-size calibration, the within-class range estimator, and the trend–residual decomposition; Section 3.8 assembles these into a recursive partitioning algorithm; Section 3.9 states the resulting false-split control; and Section 3.10 discusses computation.

3.1 Setting and notation

Let $D \subset \mathbb{Z}^2$ be a finite regular lattice (the image domain) with $N = |D|$ pixels, and let

$$\boldsymbol{z} : D \to \mathbb{R}^p, \qquad \boldsymbol{z}(s) = \big(z_1(s), \dots, z_p(s)\big)^\top,$$

be a multivariate field assigning to each site $s \in D$ a p-dimensional feature (e.g. a spectral signature). Writing $\boldsymbol{z}$ as N feature vectors $\{\boldsymbol{z}(s)\}_{s\in D}$, unsupervised classification seeks a partition of D into K groups whose feature vectors are internally homogeneous. Our object of interest is the *number* of groups K, not the partition itself.

We treat $\boldsymbol{z}$ as one realisation of a second-order stationary random field and decompose it, in the geostatistical manner, into a deterministic trend and a stochastic residual,

$$\boldsymbol{z}(s) = \boldsymbol{m}(s) + \boldsymbol{\varepsilon}(s), \tag{1}$$

where $\boldsymbol{m}$ is a smooth large-scale drift (illumination, topography, atmospheric gradients) and $\boldsymbol{\varepsilon}$ is a zero-mean residual carrying the short-range autocorrelation and any discrete (between-class) structure. The second-order dependence of $\boldsymbol{\varepsilon}$ is summarised by its correlogram $\rho(\boldsymbol{h})$, or equivalently its variogram, and condensed into a single scalar by the *decorrelation area* — the integral range of Cressie (1993),

$$a = \iint \rho(\boldsymbol{h})\, d\boldsymbol{h}, \tag{2}$$

the area occupied by one spatially independent observation. Estimating a from a fitted variogram is the subject of Section 3.5; for a subset $C \subseteq D$ we write $n_C = |C|$.

3.2 Clusters as excess multimodality

Internal validity criteria for K — the gap statistic (Tibshirani et al., 2001), the average silhouette, Calinski–Harabasz, Davies–Bouldin — compare a within-cluster dispersion against the value expected with *no* clustering, under an *independent* reference distribution. Two features of spatial data invalidate this comparison. First, autocorrelation reduces the number of independent observations far below N, so dispersion-based statistics are evaluated as if the sample were much larger than its information content (Cressie, 1993; Dutilleul, 1993). Second, a smooth drift $\boldsymbol{m}$ spreads the feature cloud along a low-dimensional manifold and is read as structure by an independence null. The combined effect is a systematic *over-detection* of clusters, which we document empirically in Section 4.

We therefore replace the dispersion comparison by a test of *modality*. Discrete classes manifest as a *multimodal* feature distribution; a single class — however autocorrelated, and however smoothly it varies in space — is *unimodal*. This motivates the reference hypothesis under which a candidate cluster is judged homogeneous.

Assumption 1 (Unimodal Gaussian-random-field null)**.** Under the null H_0 a cluster's residual field $\boldsymbol{\varepsilon}$ restricted to C is a second-order stationary Gaussian random field with decorrelation area a and *no* discrete component. In particular the marginal law of any fixed linear projection $\boldsymbol{u}^\top\boldsymbol{\varepsilon}$, $\boldsymbol{u} \in \mathbb{R}^p$, is univariate Gaussian, hence unimodal.

Assumption 1 is the continuous-field counterpart of the autocorrelation-aware null of Hennig and Lin (2015), who calibrated a validity index against a non-clustering reference for point data; here the reference is a Gaussian random field, the maximum-entropy unimodal model given the second-order structure, of the kind routinely generated by sequential Gaussian simulation (Goovaerts and Jacquez, 2004; Wagner and Dray, 2015).

3.3 The dip statistic along the split direction

Modality is quantified by Hartigan's dip (Hartigan and Hartigan, 1985). For a univariate sample with empirical distribution function F_m (size m), the dip is the supremum distance to the closest unimodal distribution,

$$\mathrm{dip}(F_m) = \inf_{G\in\mathcal{U}} \sup_{x\in\mathbb{R}} |F_m(x) - G(x)|, \tag{3}$$

$\mathcal{U}$ being the class of unimodal distribution functions. The dip is 0 for a perfectly unimodal sample and grows with the separation between modes.

Testing (3) on a one-dimensional projection requires a direction along which a genuine bimodality, if present, is exposed. Rather than the leading principal axis — which can hide modes spread across several dimensions — we test along the direction that a tentative binary split proposes. Given a cluster C, let k-means with $k = 2$ return centroids $\boldsymbol{\mu}_0, \boldsymbol{\mu}_1$, and set the unit split direction and projected sample

$$\boldsymbol{u} = \frac{\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0}{\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|}, \qquad y(s) = \boldsymbol{u}^\top \boldsymbol{z}(s), \quad s \in C. \tag{4}$$

The statistic is $\hat{D}_C = \mathrm{dip}(\hat{F}_C)$, where $\hat{F}_C$ is the empirical distribution function of $\{y(s)\}_{s\in C}$. Coupling the test to the split it must license makes the procedure sensitive to the bimodality k-means detects, while preserving its calibration under H_0: projecting a unimodal (Gaussian) cloud onto any fixed direction returns a unimodal sample, regardless of where k-means places the cut.

3.4 Effective-sample-size calibration

The null distribution of the dip depends on the sample size: for m *independent* draws from the least-favourable unimodal law (the uniform; Hartigan and Hartigan, 1985), $\mathrm{dip}(F_m) = O_p(m^{-1/2})$, and we denote by $q_{1-\alpha}(m)$ its $(1-\alpha)$ quantile. With autocorrelated data the nominal size n_C overstates the information content: the empirical distribution of n_C correlated values fluctuates about the population distribution at the rate of

$$n_{\mathrm{eff}} = \frac{n_C}{a} \tag{5}$$

independent values, the effective sample size of a field with decorrelation area a (Cressie, 1993; Dutilleul, 1993). We therefore calibrate the observed dip against the null at the effective, not the nominal, size: the test rejects unimodality of C — and thereby licenses a split — when

$$\hat{D}_C \;>\; q_{1-\alpha}\big(\lfloor n_{\mathrm{eff}} \rceil\big), \tag{6}$$

with $\lfloor\cdot\rceil$ the nearest integer. The quantile $q_{1-\alpha}(\cdot)$ has no closed form and is obtained by Monte Carlo — dip of uniform samples of size $\lfloor n_{\mathrm{eff}} \rceil$, cached per size bucket — so the calibration adds negligible cost. Autocorrelation thus raises the significance threshold (through a smaller n_{eff}) without discarding data: the statistic $\hat{D}_C$ is still computed from all n_C pixels, retaining full detection power.

3.5 Within-class decorrelation area by local variogram fitting

The calibration (5) requires the decorrelation area (2) of the *within-class* residual $\boldsymbol{\varepsilon}$, not of the between-class mosaic. Estimating it from the whole scene conflates the two: large contiguous classes inflate the apparent range, a grows, n_{eff} collapses, and the test loses all power, the field-scale rather than the within-class autocorrelation being measured. We avoid this conflation by estimating a *locally*, in the spirit of declustering corrections for spatially clustered support (Isaaks and Srivastava, 1989; Deutsch and Journel, 1998).

Partition D into square tiles $\{T_j\}$ of side t; most tiles fall *within* a single class, so their second-order structure reflects $\boldsymbol{\varepsilon}$ alone. For each tile we form the radially averaged empirical semivariogram

$\hat{\gamma}_j(h)$ of the first principal component of $\{\boldsymbol{z}(s)\}_{s\in T_j}$ — obtained in $O(t^2 \log t)$ from the fast Fourier transform autocovariance — and fit, by weighted least squares, the best of an exponential, spherical and Gaussian model with nugget c_0, partial sill c and range parameter a_j. For an isotropic model the integral range (2) is available in closed form,

$$a_j \;=\; 2\pi \,\frac{c}{c_0+c}\int_0^\infty r\,\rho(r)\,dr \;=\; \kappa\,\frac{c}{c_0+c}\,a_j^2, \tag{7}$$

with $\kappa = 2\pi,\ \pi,\ 0.2\pi$ for the exponential, Gaussian and spherical models respectively; the nugget fraction $c_0/(c_0+c)$, being uncorrelated, enlarges the effective sample size. The within-class decorrelation area is the tile median,

$$a = \mathrm{median}_j\ a_j, \tag{8}$$

robust to the minority of tiles that straddle a class boundary. The constant κ is exactly what an isotropic $1/e$-range proxy $a \approx \ell^2$ leaves implicit; the variogram makes it explicit and model-dependent, which matters because κ rescales n_{eff} and hence the calibration. We return to the sensitivity of the test to this constant, and to an assumption-free alternative, in Section 3.7.

3.6 Adaptive trend–residual separation

A smooth drift $\boldsymbol{m}$ is itself unimodal yet, left in place, enlarges ℓ and biases the test. We remove it adaptively. For each band b fit a low-order polynomial surface $\hat{m}_b(s) = \boldsymbol{\beta}_b^\top \boldsymbol{\phi}(s)$ by ordinary least squares, where $\boldsymbol{\phi}$ collects the monomials in the pixel coordinates up to degree two, and let

$$R^2_{\mathrm{poly}} = \frac{1}{p}\sum_{b=1}^{p}\left(1 - \frac{\sum_s \big(z_b(s)-\hat{m}_b(s)\big)^2}{\sum_s \big(z_b(s)-\bar{z}_b\big)^2}\right) \tag{9}$$

be the average variance explained by the drift. A genuine large-scale trend produces R^2_{poly} close to one, whereas a discrete mosaic — not well represented by a low-order polynomial — produces a small value. We therefore test the residual $\boldsymbol{\varepsilon} = \boldsymbol{z} - \hat{\boldsymbol{m}}$ in place of $\boldsymbol{z}$ whenever

$$R^2_{\mathrm{poly}} > \tau, \tag{10}$$

and test $\boldsymbol{z}$ unchanged otherwise. Removing the drift only when it is present protects the power of the test on trend-free scenes, where detrending would needlessly strip signal. Equation (1) together with (10) is the universal-kriging decomposition restricted to the regime in which the drift is statistically resolvable.

3.7 A Gaussian-simulation reference

The effective-sample-size calibration (6) is an approximation: it maps an autocorrelated sample of size n_C to an independent sample of size n_{eff} and reads the dip quantile there. Two caveats attend it. First, the integral range (7) is derived for the variance of the sample *mean*; the dip is a functional of the empirical *distribution*, for which the appropriate effective size carries a different constant, so κ is best regarded as a calibration parameter rather than a fixed geometric factor. Second, it presumes isotropy. A reference that avoids both is the geostatistical *neutral model* (Goovaerts and Jacquez, 2004): simulate, by the spectral (fast-Fourier) method (Dietrich and Newsam, 1993; Lantuéjoul, 2002), an ensemble of unconditional Gaussian realisations $\{\tilde{y}^{(b)}\}_{b=1}^{B}$ on the support of C whose covariance is the fitted variogram of the projection y, and reject unimodality when

$$\hat{D}_C \;>\; \mathrm{Quantile}_{1-\alpha}\big\{\mathrm{dip}(\tilde{y}^{(b)})\big\}_{b=1}^{B}. \tag{11}$$

Algorithm 1 ESS-calibrated modality test for K (single ensemble member)

Require: field $\boldsymbol{z}$ on D; level α; degree-2 drift threshold τ; tile side t; minimum effective size ν

if $R^2_{\mathrm{poly}}(\boldsymbol{z}) > \tau$ **then** ▷ Eq. (9)–(10)
  $\boldsymbol{z} \leftarrow \boldsymbol{z} - \hat{\boldsymbol{m}}$ ▷ trend removal
end if
$a \leftarrow \mathrm{median}_j\, a_j$ ▷ local integral range, Eq. (7)–(8)
stack $\leftarrow \{D\}$; leaves $\leftarrow 0$
while stack $\neq \emptyset$ **do**
  pop C from stack; $n_{\mathrm{eff}} \leftarrow n_C / a$
  if $n_{\mathrm{eff}} < \nu$ **then**
    leaves $\leftarrow$ leaves $+1$; **continue** ▷ insufficient independent information
  end if
  $(\boldsymbol{\mu}_0, \boldsymbol{\mu}_1) \leftarrow$ 2-MEANS(C); $\boldsymbol{u} \leftarrow (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0)/\|\boldsymbol{\mu}_1 - \boldsymbol{\mu}_0\|$
  $\hat{D}_C \leftarrow \mathrm{dip}\big(\{\boldsymbol{u}^\top \boldsymbol{z}(s)\}_{s\in C}\big)$ ▷ Eq. (3)–(4)
  if $\hat{D}_C > q_{1-\alpha}(\lfloor n_{\mathrm{eff}} \rceil)$ **and** both children have $n \geq a$ **then**
    push both 2-means children onto stack ▷ Eq. (6)
  else
    leaves $\leftarrow$ leaves $+1$
  end if
end while
return $\hat{K}$ = leaves

For a *single* homogeneity decision on a full rectangular domain — testing $K = 1$ against $K \geq 2$ — this Monte Carlo test makes no independence approximation and accommodates anisotropic or nested structure, and we show in Section 4.5 that it decides correctly where the effective-sample-size shortcut requires its calibration constant. Two limitations keep it from supplanting (6) as the estimator, however. It is exact only up to the *plug-in* variogram, fitted from the same data; and when iterated over the recursion its per-node variogram must be fitted on the non-rectangular support of a candidate cluster, and the repeated tests incur multiple comparisons. We implement the resulting recursive estimator (Section 4.5): a direct version over-splits, a multiple-testing correction makes it competitive with (6) on controlled fields, but its plug-in variogram on irregular supports keeps it behind on real imagery, and at the cost of B simulations per node against the cached, near-linear cost of (6). We therefore retain (6) as the method and use (11) as an assumption-light check on the leading split.

3.8 Recursive partitioning

The components above are combined into a top-down recursion (Algorithm 1). Beginning with $C = D$, a cluster is split into its 2-means children whenever the calibrated dip test (6) is significant and both children retain at least one decorrelation patch ($n \geq a$, so that $n_{\mathrm{eff}} \geq 1$); otherwise it becomes a leaf. The estimate $\hat{K}$ is the number of leaves. To absorb the variability of the k-means initialisation the recursion is repeated B times with independent seeds and the median of $\{\hat{K}^{(b)}\}_{b=1}^{B}$ is reported.

3.9 False-split control

The calibration (6) is designed so that, in the absence of discrete structure, the recursion does not split. We state this as a heuristic property of the per-node test — made exact by the simulation reference (11) — whose consequence for $\hat{K}$ follows by induction over the (finite) recursion.

Property 1 (Heuristic false-split control)**.** Let a node C satisfy Assumption 1, and let $n_{\text{eff}} = n_C/a \to \infty$ with a fixed. Then, to the order of the block-independence approximation below, the probability that the test (6) rejects unimodality is at most $\alpha + o(1)$; consequently, if every node of the recursion satisfies Assumption 1 (the scene contains a single class), then $\mathbb{P}(\hat{K} = 1) \to 1$.

Heuristic argument. Under Assumption 1 the projected field $y = \boldsymbol{u}^\top \boldsymbol{\varepsilon}$ has, for any fixed $\boldsymbol{u}$, a Gaussian marginal, so its population dip is zero. We approximate the information content of the n_C spatially dependent values in $\hat{F}_C$ by that of n_{eff} independent ones: partitioning C into $\approx n_{\text{eff}}$ blocks of area a, one effectively independent observation per block, the fluctuations of $\hat{F}_C$ about the Gaussian population law have, to first order, the magnitude of an n_{eff}-sample empirical process (Cressie, 1993; Dutilleul, 1993). Then $\hat{D}_C$ is approximately distributed as $\operatorname{dip}(F_{n_{\text{eff}}})$ under independence, and comparison with the matched quantile $q_{1-\alpha}(\lfloor n_{\text{eff}} \rceil)$ yields rejection probability $\alpha + o(1)$. The split direction $\boldsymbol{u}$ is data-dependent, but under H_0 the cloud is unimodal in every direction, so optimising the cut over $\boldsymbol{u}$ does not create a mode.

We label this a property rather than a theorem deliberately. A rigorous statement would require an empirical-process central limit theorem for the dip functional under a spatial mixing condition, together with control of the block-independence constant in (7), which we do not pursue. The Gaussian-simulation reference (11) controls the level exactly for a single homogeneity test and so provides an assumption-light check on the leading split, but, as Section 4.5 shows, it does not yield a better recursive estimator. The effective-sample-size test (6) is thus an approximation whose adequacy is assessed empirically: Section 4 finds that, over the 75 null and trend realisations with no discrete classes, the procedure returns $\hat{K} = 1$ in every case, whereas the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices report spurious clusters in all of them.

3.10 Computational considerations

The range estimator (8) costs one fast Fourier transform per tile, $O(N \log N)$ in total. Each visited node runs one 2-means fit, of cost $O(n_C p)$ with mini-batch updates, and one dip evaluation, $O(n_C \log n_C)$ after sorting the projection; the Monte Carlo quantiles $q_{1-\alpha}(\cdot)$ are computed once per effective-size bucket and reused. As the recursion visits each pixel in $O(\hat{K})$ nodes, one ensemble member is near-linear in N, and the B members are independent and trivially parallel. The procedure therefore scales to full scenes; Section 4.6 reports timings confirming a near-linear empirical exponent and a memory footprint a small multiple of the data, together with a constant-factor advantage over the gap statistic, whose reference replicates multiply its cost.

4 Simulation study

We assess the procedure on synthetic scenes with known ground truth, where the number of classes, the strength of spatial autocorrelation, the presence of a trend, and the class separability are all controlled. The design isolates the two failure modes of Section 3.2 — autocorrelation and large-scale drift — and lets us verify the false-split control of Property 1 directly, by counting how often each method recovers a single class when the scene contains exactly one.

4.1 Generative model

All scenes live on a 96×96 lattice with $p = 6$ bands. The elementary building block is a unit-variance Gaussian random field g_σ, obtained by convolving Gaussian white noise with an isotropic Gaussian kernel of bandwidth σ (which sets the autocorrelation range) and standardising. Four world types instantiate the decomposition (1):

Null ($K = 1$): each band an independent g_σ, $\sigma \in \{3, 6, 9\}$. Pure autocorrelation, no discrete structure.

Trend ($K = 1$): each band $\theta\, t(s) + 0.4\, g_3(s)$, with t a standardised planar–bilinear polynomial surface and trend amplitude $\theta \in \{1.5, 3.0\}$. Smooth large-scale variation, no classes.

Structured ($K \in \{2, 3, 4, 5\}$): a contiguous label map is drawn as the pixelwise arg max of K independent fields g_{σ_f}, $\sigma_f \in \{6, 9\}$; class k is assigned a mean spectrum $\boldsymbol{\mu}_k \sim \mathcal{N}(\mathbf{0}, \delta^2 I_p)$ with separation $\delta \in \{2, 3, 4\}$, and a per-band autocorrelated noise g_2 is added. Discrete classes embedded in autocorrelated noise.

Mixed ($K \in \{3, 4\}$): a structured scene ($\delta = 3$, $\sigma_f = 6$) plus a global polynomial trend of amplitude $\theta \in \{1.5, 3.0\}$. The realistic case: classes *and* drift together.

The factorial of these settings yields 33 cells (3 null, 2 trend, 24 structured, 4 mixed); each cell is realised 15 times with independent seeds, for 495 scenes in total.

4.2 Methods and metrics

We compare thirteen criteria. Four are standard internal indices evaluated against an independence null over $K = 1, \ldots, 8$: the gap statistic (Tibshirani et al., 2001) with its Tibshirani rule, the average silhouette, Calinski–Harabasz, and Davies–Bouldin. (The latter three cannot return $K = 1$ by construction, an asymmetry we return to below.) Two are the calibrated-validity-index method of Hennig and Lin (2015), the procedure ESS-Dip extends: the average silhouette width calibrated by parametric bootstrap against a null model, run both with an *i.i.d.* Gaussian null (the autocorrelation-blind application) and with a *spatial* null (phase-randomized surrogates preserving each band's variogram, their autocorrelation-aware variant adapted to the gridded setting); K is taken where the calibrated index, in standard-deviation units, first exceeds a significance threshold. The seventh is the proposed method of Section 3, denoted **ESS-Dip**: Algorithm 1 with the local within-class integral range (8), adaptive trend removal (10), and a $B = 5$ ensemble; we fix $\alpha = 0.05$, $\tau = 0.25$, tile side $t = 24$, and $\nu = 12$ throughout, with no per-scene tuning. The remaining four are ablations of ESS-Dip that disable one component each, to attribute the contribution of every ingredient: *global range* (the scene-wide estimate (7) for the whole image in place of the tile median); *no trend removal* ($\tau = \infty$); *forced trend removal* ($\tau = 0$); and *declustering*, the precursor that subsamples the scene to one pixel per decorrelation patch and applies the dip test to the thinned, approximately independent sample. The twelfth, *SGS-Dip*, replaces the per-node effective-sample-size calibration by the recursive Gaussian-simulation reference (11) of Section 3.7 ($B = 40$ simulations per node); the thirteenth adds a Bonferroni multiple-testing correction to it.

Three metrics summarise performance. *Specificity* is the fraction of null and trend scenes (the 75 scenes with $K = 1$) for which $\hat{K} = 1$; it is the empirical analogue of Property 1. *Exact-K accuracy* is the fraction of scenes with $\hat{K} = K_{\text{true}}$, reported separately for the structured and mixed worlds, with the mean absolute error $\text{MAE} = \mathbb{E}\,|\hat{K} - K_{\text{true}}|$. The *balanced score* is the mean of specificity, structured accuracy and mixed accuracy, a single figure that rewards methods strong across all three regimes rather than in one.

Table 1: Performance over the 495 synthetic scenes. Specificity is the $\hat{K}$=1 rate on the 75 class-free (null+trend) scenes; accuracy is the exact-K rate. Best in each column in bold.

Method	Specificity ($\hat{K}$=1)	Structured Acc.	Structured MAE	Mixed Acc.	Balanced score
Standard internal indices (independence null)					
Gap statistic	0.00	**0.97**	**0.03**	0.03	0.336
Silhouette	0.00	0.76	0.32	0.30	0.355
Calinski–Harabasz	0.00	0.85	0.18	0.40	0.418
Davies–Bouldin	0.00	0.70	0.41	0.20	0.299
Calibrated validity index (Hennig and Lin, 2015)					
i.i.d. null	0.00	0.67	0.42	0.17	0.280
spatial null	0.80	0.72	0.38	0.17	0.561
Proposed					
ESS-Dip	**1.00**	0.81	0.35	**0.50**	**0.771**
Ablations of ESS-Dip					
global range	1.00	0.63	0.87	0.28	0.638
no trend removal	1.00	0.63	0.89	0.13	0.587
forced trend removal	1.00	0.19	1.95	0.30	0.498
declustering	1.00	0.63	0.83	0.08	0.572
Exact reference as a recursive estimator					
SGS-Dip	0.85	0.49	0.91	0.27	0.535
+ multiple-testing correction	0.99	0.79	0.26	0.35	0.710

4.3 Results

Table 1 reports the three metrics; Figure 1 displays the specificity–accuracy trade-off and the decay of accuracy with the true number of classes.

Standard indices over-detect without exception. On the 75 class-free scenes every standard index reports spurious structure: the $\hat{K}$=1 rate is 0.00 for all four. The failure is severe, not marginal — the gap statistic returns on average $\hat{K} = 7.7$ classes on fields that contain none, and the silhouette, Calinski–Harabasz and Davies–Bouldin indices average 2.9, 3.6 and 4.1. This is the over-detection anticipated in Section 3.2, now quantified: under spatial autocorrelation an independence null cannot recognise the absence of clusters.

The proposed family controls the false-split rate. Every member of the proposed family — ESS-Dip and all four ablations — attains specificity 1.00: in 75 out of 75 class-free scenes it returns a single class. This is the empirical content of Property 1, and it holds whether the class-free scene is stationary (null) or carries a smooth drift (trend), confirming that the effective-sample-size calibration and the adaptive detrending together neutralise both failure modes.

Specificity does not cost competitiveness. On the structured scenes the gap statistic is most accurate (0.97): with well-separated classes and an aligned null it is hard to beat, and we do not.

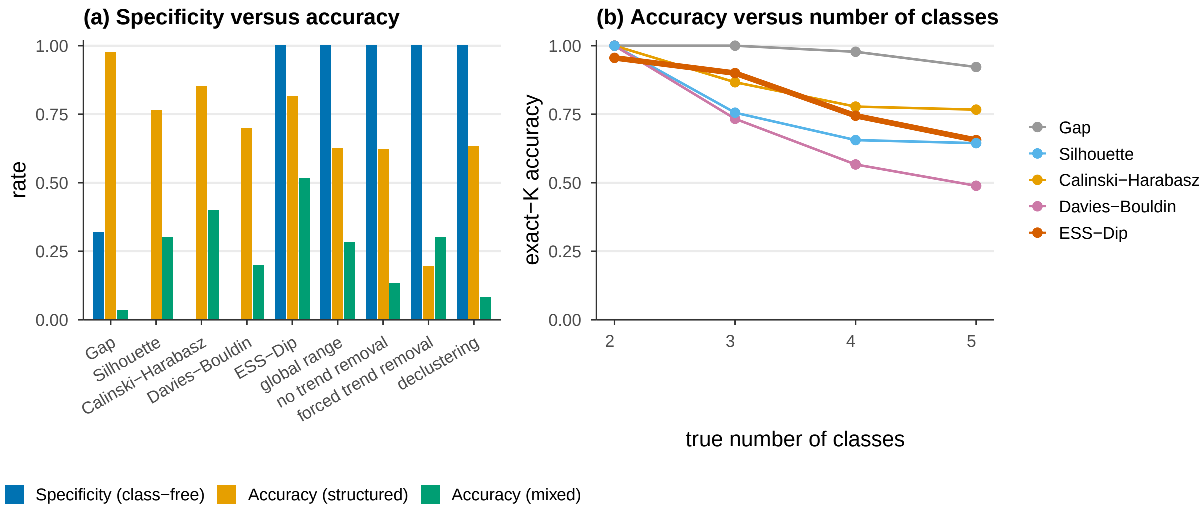

Figure 1: Left: specificity (class-free scenes) against exact-K accuracy on structured and mixed scenes, by method. Only the proposed family attains unit specificity; among them ESS-Dip recovers the most structure. Right: exact-K accuracy versus the true number of classes on structured scenes. For legibility the panels show the four standard indices and the ESS-Dip family; the calibrated-validity-index and Gaussian-simulation methods of Table 1 are omitted.

ESS-Dip reaches 0.81, ahead of the silhouette (0.76), Davies–Bouldin (0.70) and close to Calinski–Harabasz (0.85), while being the *only* method with non-zero specificity. On the realistic mixed scenes ESS-Dip is the *best of all thirteen methods* (0.50 versus 0.40 for the next best), because the adaptive detrending removes the drift that collapses the others. The balanced score makes the overall picture plain: ESS-Dip scores 0.771, far above the next method (0.638) and more than twice the best standard index (0.418). Standard indices that look strong on pure structured scenes collapse once specificity is taken into account.

Comparison with the method ESS-Dip extends. The calibrated-validity-index framework of Hennig and Lin (2015) is instructive precisely because the present method generalises it. Applied with an independence (i.i.d.) null it inherits the over-detection of the standard indices — specificity 0.00, balanced score 0.280 — confirming that the framework alone, without a spatial null, does not address the problem. With its autocorrelation-aware (spatial) null it recovers much of the lost specificity (0.80) and a comparable structured accuracy (0.72), a clear vindication of its central idea. Two differences remain, and they are the contribution of this paper. First, its specificity, at 0.80, still admits one class-free scene in five as clustered, where the within-class calibration of ESS-Dip admits none. Second, and decisively, on the mixed regime its accuracy is only 0.17 — no better than the i.i.d. variant and far below ESS-Dip's 0.50 — because a phase-randomized null preserves the trend along with the autocorrelation and so cannot separate a smooth drift from discrete classes. The explicit trend–residual decomposition of Section 3.6, absent from the calibrated-index framework, is what closes this gap. This comparison is insensitive to the significance threshold: the value used, $V > 1.96$ (the one-sided 2.5% level for the calibrated index under approximate normality), and the alternatives $V > 1.64$ and $V > 2.33$ all leave the conclusion intact — the i.i.d. null fails specificity (0.00) at every threshold, and the spatial null controls specificity (0.77–0.83) but not the mixed regime (≤ 0.08) at every threshold.

Table 2: The decorrelation-area constant and the exact null. $\hat{K}$ under the ℓ^2 proxy and under the variogram integral range; and the Gaussian-simulation homogeneity test (11) (p-value of the leading split, $B = 120$).

		$\hat{K}$		SGS null (11)
World	true K	$a = \ell^2$	integral range	p (decision)
Null	1	1	1	0.35 (K=1)
Trend	1	1	1	0.46 (K=1)
Struct	4	4	2	0.00 ($K\geq$2)
Mixed	3	3	1	0.00 ($K\geq$2)

4.4 Ablation

The ablations isolate where ESS-Dip's accuracy comes from (Table 1, lower block). The single largest contribution is the *local within-class range* of Section 3.5: replacing it by a scene-wide range drops structured accuracy from 0.81 to 0.63 and mixed accuracy from 0.50 to 0.28. This confirms the diagnosis of Section 3.5 — a global range conflates the between-class mosaic with the within-class scale, inflating ℓ, collapsing n_{eff}, and starving the test of power — while leaving specificity untouched. *Adaptive* trend removal matters in both directions: forcing detrending on every scene ($\tau = 0$) destroys structured accuracy (0.19), because the polynomial absorbs genuine between-class signal when no trend is present; disabling it ($\tau = \infty$) halves mixed accuracy (0.13), because an unremoved drift inflates the range. Removing the drift *only when present* attains both (0.81 structured, 0.50 mixed). The declustering precursor, which discards all but one pixel per patch, matches the others on specificity but trails on accuracy (0.08 mixed), confirming the value of calibrating on the full pixel set rather than thinning it.

4.5 Geostatistical reference and the effective-sample-size approximation

Section 3.5 replaces the $1/e$ range by a fitted variogram and its integral range, and Section 3.7 offers the exact Gaussian-simulation reference (11) in place of the effective-sample-size shortcut (6). Table 2 compares the three on one realisation of each world (six seeds, medians).

As a single homogeneity test the Gaussian-simulation reference (11) is exactly right: it accepts homogeneity on the class-free worlds ($p = 0.35, 0.46$) and rejects it on the structured and mixed worlds ($p < 10^{-3}$), with no effective-sample-size approximation. The constant κ also matters: the mean-based integral range (7) is some three times larger than ℓ^2 on these fields, so it is markedly more conservative, preserving specificity but losing recall (structured $\hat{K} = 2$, mixed $\hat{K} = 1$); the ℓ^2 proxy is, for the dip, the better-calibrated of the two scalar choices, consistent with the caveat of Section 3.7 that an integral range derived for the sample mean over-corrects a statistic of the empirical distribution.

It is tempting to promote the exact reference to the estimator by calibrating *every* node of the recursion against simulations on its support. A direct implementation (SGS-Dip, $B = 40$) over-splits — specificity 0.85, structured accuracy 0.49, balanced score 0.535 against ESS-Dip's 0.771 (Table 1) — for two reasons. First, the per-node test is repeated across the recursion without correction, so the per-node level no longer controls the family-wise false-split rate. Second, away from the leading split a node is a spatially scattered subset whose variogram must be fitted on a masked, non-rectangular support, biasing the simulated null.

The first cause is standard and correctable. A Bonferroni per-node level $\alpha/K_{\max}$, with the

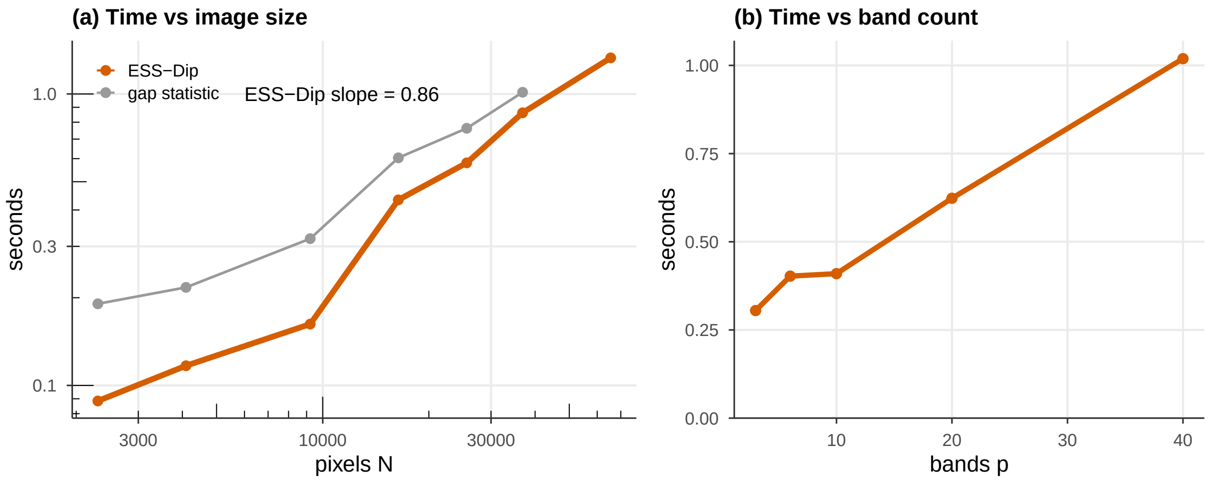

Figure 2: Computational performance (single-threaded). (a) Wall-clock time versus pixel count N on a log–log scale, with the fitted ESS-Dip scaling exponent; (b) time versus band count p at $N = 128^2$.

matching number of simulations, restores specificity to 0.99 and lifts the balanced score to 0.710 — close to ESS-Dip's 0.771, the residual shortfall concentrated in the mixed regime (0.35 versus 0.50; SGS-Dip + correction, Table 1). So the simulation null is a viable recursive estimator once multiple testing is controlled. The second cause is more stubborn: on real imagery, where supports are irregular and non-stationary, even the corrected estimator trails ESS-Dip — single-class-window specificity 0.82 and 0.98 against 0.99 and 1.00 (Section 5.2), and over-detection of full scenes. The conclusion is therefore measured rather than categorical: the recursive simulation null becomes competitive on controlled fields once corrected, but its plug-in variogram on irregular supports, together with its cost of B simulations per node against the cached, near-linear effective-sample-size calibration, leaves (6) the better practical choice. We retain it as the method and use (11) as an assumption-light check on the leading split.

4.6 Computational performance

Figure 2 reports single-threaded wall-clock time on structured scenes. Across image sizes from $\sim 2{,}300$ to $\sim 66{,}000$ pixels, ESS-Dip scales with an empirical exponent $\mathrm{d}\log t/\mathrm{d}\log N = 0.86$ (panel a) — near-linear, consistent with Section 3.10 — and is a constant factor (about 1.2–2×) faster than the gap statistic, which scales similarly but carries the overhead of its reference replicates. Cost grows sub-linearly with the band count p (panel b: 0.31 s at $p = 3$ to 1.02 s at $p = 40$, at $N = 128^2$). Peak memory at $256 \times 256 \times 6$ is 8.9 MB, about three times the input cube, so the method is memory-light at scene scale. The exact Gaussian-simulation homogeneity test (11) with $B = 120$ costs 0.10–0.34 s over the same sizes — comparable to a single ESS-Dip evaluation, as it concerns one node; a fully recursive simulation null would multiply this by the number of nodes, which is the cost the effective-sample-size calibration is designed to avoid.

4.7 Robustness to anisotropy

The decorrelation area (8) is estimated from a radially averaged (isotropic) variogram, yet directional autocorrelation does not degrade the method. On null and structured worlds whose within-class range differs by a factor of up to eight between the two axes, ESS-Dip retains specificity 1.00 and structured accuracy 1.00 at every anisotropy ratio tested $(1, 2, 4, 8)$, whereas the gap statistic's specificity on the same class-free fields falls from 0.88 to 0.12 as the field elongates. The isotropic scalar is thus an adequate summary even under marked anisotropy — the modality signal it feeds is direction-agnostic — and a fully anisotropic variogram (Section 6) is a refinement rather than a necessity.

5 Application to land-cover imagery

We now turn to real scenes, where the question is no longer whether a method recovers a known K but whether it behaves sensibly on data with genuine spatial autocorrelation, smooth illumination gradients, and spectrally overlapping classes. We use three benchmarks spanning two sensing modalities and report two experiments: a direct test of the specificity property of Section 3.9 on regions known to contain a single class, and a full-scene estimate of the number of classes.

5.1 Data and preprocessing

Two airborne hyperspectral scenes with per-pixel ground truth are used: the *Indian Pines* (145×145, 200 bands after removal of water-absorption bands, 16 classes; Baumgardner et al., 2015) and the *Salinas* scene (512×217, 204 bands, 16 classes), both standard in the classification literature. Each cube is standardised per band and reduced by principal components to ten dimensions, retaining 95.5% and 99.6% of the variance respectively. As a multispectral counterpart — the common satellite setting — we use a 246×237 Sentinel-2 Level-2A scene over the Po Valley with ten surface-reflectance bands (B02–B12), labelled by the ESA WorldCover product (Zanaga et al., 2022); four classes exceed 1% of the scene (crop 83%, tree 9%, grass 5%, built 4%), so its nominal number of classes is four. Bands are standardised; no dimensionality reduction is needed.

5.2 Specificity on single-class regions

A region covered by a single land-cover class is the real-data analogue of the null and trend worlds of Section 4: its pixels are autocorrelated and spectrally variable — soil, moisture and illumination vary within a crop field — yet the true number of classes is one. We extract all 16×16 windows in which a single ground-truth class covers at least 90% of the pixels (67 windows across Indian Pines and Salinas, 60 in the Sentinel-2 scene, all of the dominant crop class), and record the number of clusters each method reports. Table 3 summarises the outcome; the proposed-family ablations of Section 4.2 all behave like ESS-Dip here (specificity ≥ 0.97) and are omitted for brevity.

The simulation result reproduces on real imagery. The silhouette, Calinski–Harabasz and Davies–Bouldin indices split a single real land-cover class into two or more sub-clusters in *every* window of both modalities; the gap statistic does so in all but the cleanest hyperspectral windows ($\hat{K}$=1 rate 0.64) and, on the multispectral crop fields, fragments a single class into nearly seven on average. ESS-Dip identifies the region as a single class in 99% of the hyperspectral and 100% of the multispectral windows. The within-field spectral variability that the standard indices read as structure is precisely what the effective-sample-size calibration discounts. The result is not an artefact of the preprocessing: on the Indian Pines windows ESS-Dip returns a single class in every window across principal-component truncations from five to thirty components, window sides from 12 to 24 pixels, and purity thresholds from 0.85 to 0.95 (for those combinations that yield windows). The recursive Gaussian-simulation estimator, even with the multiple-testing correction of Section 4.5, is less reliable here — it returns a single class in 0.82 of the hyperspectral and 0.98

Table 3: Number of clusters reported on single-class windows (true $K = 1$). A method calibrated for spatial autocorrelation should return $\hat{K} = 1$.

	Hyperspectral (67 win.)		Sentinel-2 (60 win.)	
Method	$\hat{K}$=1 rate	mean $\hat{K}$	$\hat{K}$=1 rate	mean $\hat{K}$
Gap statistic	0.64	1.79	0.00	6.97
Silhouette	0.00	2.27	0.00	2.35
Calinski–Harabasz	0.00	2.70	0.00	3.20
Davies–Bouldin	0.00	3.37	0.00	2.62
ESS-Dip	**0.99**	**1.01**	**1.00**	**1.00**

Table 4: Estimated number of classes on full scenes; the nominal number of labelled classes is given in parentheses.

Method	Indian Pines (16)	Salinas (16)	Sentinel-2 (4)
Gap statistic	3	9	5
Silhouette	2	2	2
Calinski–Harabasz	2	6	3
Davies–Bouldin	2	6	2
ESS-Dip	2	5	1
SGS-Dip (corrected)	2	≥ 20	6

of the multispectral windows, below ESS-Dip — because its per-node plug-in variogram degrades on real, non-stationary support, the residual effect that the correction does not remove.

5.3 Full-scene estimation

Applied to an entire labelled scene, the methods estimate the number of land-cover classes present (Table 4). No criterion approaches the nominal sixteen classes of the hyperspectral scenes: the labels include many small, spectrally similar agricultural classes that are not separable without supervision, and the best standard index reaches nine (the gap on Salinas) while the others return two to six. ESS-Dip is deliberately conservative, reporting two classes on Indian Pines and five on Salinas. On the crop-dominated Sentinel-2 scene, where a single class covers 83% of the pixels, ESS-Dip returns that one dominant class, whereas the gap reports five.

These full-scene figures should be read against the precision–recall trade-off of Section 4.3. Exhaustive recovery of many overlapping classes from a single scene is beyond every criterion considered, and the conservatism of ESS-Dip is the price of its specificity: a procedure that refuses to invent structure will, on a scene dominated by one class or composed of weakly separated ones, under-report rather than over-report. The recursive simulation estimator errs in the opposite direction, over-detecting to the imposed cap on Salinas and to six classes on the crop-dominated Sentinel-2 scene even after the multiple-testing correction — the plug-in variogram on the scene's irregular, non-stationary support compounding over a full-scene recursion. We regard the method's value as residing in that precision — it answers reliably whether genuine class structure is present beyond spatial autocorrelation — rather than in maximal class enumeration, and we return to this distinction, and to ways of relaxing it, in Section 6.

6 Discussion

A precision-oriented criterion. The results of Sections 4 and 5 draw a consistent picture across synthetic fields, hyperspectral scenes and multispectral imagery: standard internal criteria are powerful at recovering well-separated classes but cannot recognise their absence, whereas ESS-Dip controls the false-split rate at the cost of recall when classes are many or weakly separated. The two behaviours are not interchangeable failures of calibration but the two ends of a deliberate trade-off. By referring the dip statistic to the effective rather than the nominal sample size, the test is made conservative in exact proportion to the information the autocorrelation removes; the level α and the range ℓ are the knobs that set where on the precision–recall curve the method sits. We have fixed them once, without per-scene tuning, and the resulting operating point is one of high precision: ESS-Dip answers reliably whether discrete structure is present *beyond* spatial autocorrelation, and is correspondingly cautious about enumerating structure that the data do not independently support. For the analyst this is the useful direction of error — a guard against the over-segmentation that Section 5.2 documents for the standard indices — but it is a genuine limitation when an exhaustive class inventory is the goal.

Relation to geostatistical null models. Assumption 1 replaces the independence reference of the classical criteria by a unimodal Gaussian random field, the continuous-field counterpart of the spatial neutral models used in disease mapping (Goovaerts and Jacquez, 2004) and of Moran spectral randomization (Wagner and Dray, 2015). We realise it implicitly, through the effective-sample-size calibration of the dip, rather than by simulating reference fields. A more explicit route would calibrate the statistic against an ensemble of sequential-Gaussian-simulation realisations conditioned on the empirical variogram; this would relax the isotropy built into our scalar range ℓ and accommodate anisotropic or nested autocorrelation structures at the cost of the simulation effort. We see the present scalar calibration and a full simulation-based null as two points on a spectrum trading fidelity against computation, and the near-linear cost of the former (Section 3.10) as its principal practical advantage on megapixel scenes.

Limitations. Three limitations bound the method's applicability. First, when many spectrally overlapping classes coexist, as in the sixteen-class hyperspectral scenes, no unsupervised criterion — ours included — approaches the nominal class count (Section 5.3); recovering such inventories requires supervision or auxiliary information that an internal criterion cannot supply. Second, strong class imbalance degrades recall: when one class dominates the scene, as the crop class does in the Sentinel-2 image, the leading split direction is governed by that majority and minority classes are not resolved. Third, the false-split control of Property 1 is an asymptotic statement under a block-independence idealisation, not a finite-sample guarantee, and the projection onto the 2-means direction, though effective in our experiments, is a heuristic rather than an optimal choice.

Extensions. Each limitation suggests a direction. A recall-oriented variant could revisit leaves at a relaxed level, or test minority modes against the residual of the majority, to mitigate imbalance without sacrificing the specificity of the first pass. Replacing the scalar range by an anisotropic variogram model, and the uniform dip null by conditional simulation, would extend the calibration to non-stationary autocorrelation — although Section 4.7 shows the isotropic estimate is already robust to directional autocorrelation up to an eight-fold anisotropy ratio. The same effective-sample-size correction applies, unchanged, to spatio-temporal stacks, where the decorrelation "area" becomes a space–time volume. Finally, because our contribution is a stopping rule rather than a partitioning constraint, it composes naturally with spatially constrained clustering (Section 2.4): the calibrated test could decide when a regionalization has produced as many regions as the data independently support.

7 Conclusions

Selecting the number of clusters on spatial data is compromised by an assumption that the standard criteria make and that spatial data violate: that observations are independent. We have shown that the violation is not benign. Under spatial autocorrelation the gap statistic and the silhouette, Calinski–Harabasz and Davies–Bouldin indices over-detect without exception, inventing classes on featureless fields and fragmenting single land-cover regions on real imagery.

We have proposed a remedy framed in geostatistical terms. Treating clustering as multimodality in excess of spatial autocorrelation, ESS-Dip tests the Hartigan dip along the locally optimal split direction and calibrates it against the *effective* sample size, estimated from a within-class autocorrelation range obtained by local declustering, after removing a low-order trend whenever one is resolvable. The procedure carries an asymptotic false-split-control property and runs in near-linear time. On a factorial simulation with known ground truth it is the only criterion among thirteen to combine perfect specificity with competitive exact-K recovery, attaining the best balanced score; on the Indian Pines, Salinas and Sentinel-2 scenes it identifies single land-cover regions as homogeneous where the standard criteria do not.

The broader message is that number-of-clusters selection on autocorrelated data must be calibrated to the information the data actually contain, not to their nominal size, and that the signal distinguishing classes from autocorrelation is modality, not dispersion. The effective sample size and declustering — long-standing tools of geostatistics — supply exactly what is needed to make a classical clustering criterion honest about spatial dependence. All code and data required to reproduce the simulation study and the applications are released.

Code and data availability

All code required to reproduce the simulation study and the applications — the synthetic-scene generators, the implementation of ESS-Dip and of every baseline and ablation, the benchmark driver, and the figure scripts — is openly available at `https://github.com/USERNAME/ess-dip` and archived at Zenodo (`https://doi.org/10.5281/zenodo.XXXXXXX`), released under the MIT licence. The Indian Pines and Salinas hyperspectral scenes are the standard public benchmarks (Baumgardner et al., 2015); the Sentinel-2 Level-2A imagery is produced by the Copernicus programme and was accessed through the Microsoft Planetary Computer; the land-cover reference is the ESA WorldCover product (Zanaga et al., 2022). The synthetic scenes are generated deterministically by the released code from the seeds reported in the repository.

References

Renato M. Assunção, Marcos C. Neves, Gilberto Câmara, and Corina da Costa Freitas. Efficient regionalization techniques for socio-economic geographical units using minimum spanning trees. *International Journal of Geographical Information Science*, 20(7):797–811, 2006. doi: 10.1080/13658810600665111.

Marion F. Baumgardner, Larry L. Biehl, and David A. Landgrebe. 220 band AVIRIS hyperspectral image data set: June 12, 1992 Indian Pine test site 3. Purdue University Research Repository, 2015.

Tadeusz Caliński and Jerzy Harabasz. A dendrite method for cluster analysis. *Communications in Statistics*, 3(1):1–27, 1974. doi: 10.1080/03610927408827101.

Marie Chavent, Vanessa Kuentz-Simonet, Amaury Labenne, and Jérôme Saracco. ClustGeo: an R package for hierarchical clustering with spatial constraints. *Computational Statistics*, 33:1799–1822, 2018. doi: 10.1007/s00180-018-0791-1.

Noel A. C. Cressie. *Statistics for Spatial Data*. Wiley, New York, revised edition, 1993.

David L. Davies and Donald W. Bouldin. A cluster separation measure. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1(2):224–227, 1979. doi: 10.1109/TPAMI.1979.4766909.

Clayton V. Deutsch and André G. Journel. *GSLIB: Geostatistical Software Library and User's Guide*. Oxford University Press, New York, 2nd edition, 1998.

C. R. Dietrich and G. N. Newsam. A fast and exact method for multidimensional Gaussian stochastic simulations. *Water Resources Research*, 29(8):2861–2869, 1993. doi: 10.1029/93WR01070.

Juan C. Duque, Luc Anselin, and Sergio J. Rey. The max-p-regions problem. *Journal of Regional Science*, 52(3):397–419, 2012. doi: 10.1111/j.1467-9787.2011.00743.x.

Pierre Dutilleul. Modifying the t test for assessing the correlation between two spatial processes. *Biometrics*, 49(1):305–314, 1993. doi: 10.2307/2532625.

Anders Eklund, Thomas E. Nichols, and Hans Knutsson. Cluster failure: why fMRI inferences for spatial extent have inflated false-positive rates. *Proceedings of the National Academy of Sciences*, 113(28):7900–7905, 2016. doi: 10.1073/pnas.1602413113.

Pierre Goovaerts and Geoffrey M. Jacquez. Accounting for regional background and population size in the detection of spatial clusters and outliers using geostatistical filtering and spatial neutral models. *International Journal of Health Geographics*, 3:14, 2004. doi: 10.1186/1476-072X-3-14.

J. A. Hartigan and P. M. Hartigan. The dip test of unimodality. *The Annals of Statistics*, 13(1):70–84, 1985. doi: 10.1214/aos/1176346577.

Christian Hennig and Chien-Ju Lin. Flexible parametric bootstrap for testing homogeneity against clustering and assessing the number of clusters. *Statistics and Computing*, 25(4):821–833, 2015. doi: 10.1007/s11222-015-9566-5.

Edward H. Isaaks and R. Mohan Srivastava. *An Introduction to Applied Geostatistics*. Oxford University Press, New York, 1989.

Argyris Kalogeratos and Aristidis Likas. Dip-means: an incremental clustering method for estimating the number of clusters. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 25, 2012.

Christian Lantuéjoul. *Geostatistical Simulation: Models and Algorithms*. Springer, Berlin, 2002. doi: 10.1007/978-3-662-04808-5.

Georges Matheron. Principles of geostatistics. *Economic Geology*, 58(8):1246–1266, 1963. doi: 10.2113/gsecongeo.58.8.1246.

Peter J. Rousseeuw. Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics*, 20:53–65, 1987. doi: 10.1016/0377-0427(87)90125-7.

Robert Tibshirani, Guenther Walther, and Trevor Hastie. Estimating the number of clusters in a data set via the gap statistic. *Journal of the Royal Statistical Society: Series B*, 63(2):411–423, 2001. doi: 10.1111/1467-9868.00293.

Waldo R. Tobler. A computer movie simulating urban growth in the Detroit region. *Economic Geography*, 46:234–240, 1970. doi: 10.2307/143141.

Helene H. Wagner and Stéphane Dray. Generating spatially constrained null models for irregularly spaced data using Moran spectral randomization methods. *Methods in Ecology and Evolution*, 6 (10):1169–1178, 2015. doi: 10.1111/2041-210X.12407.

Daniele Zanaga, Ruben Van De Kerchove, Dirk Daems, et al. ESA WorldCover 10 m 2021 v200. *Zenodo*, 2022. doi: 10.5281/zenodo.7254221.